Notes on MRI Distortions

MRI work group

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Here is a first collection of distortions that could be relevant for implementation in scope of the MRI Metrics project (activity related to Metrics Reloaded). The most relevant distortions are being implemented in `https://github.com/MatthiasLen/mri_recon` alongside a collection of classic reconstruction algorithms, wrappers to pretrained NN approaches and a set of well-established image metrics.

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2D Cartesian k-space distortions for MRI reconstruction testing

Below are k-space distortions tailored for 2D Cartesian MRI (kx: readout/frequency-encode, ky: phase-encode). For each item: what to do in k-space, the key parameter(s) to sweep, and the expected image-domain effect.

Assumed Fourier convention

We adopt the following Fourier pair and sign convention:

- Forward: $M(k_x, k_y) = \iint m(x, y)\, e^{-i2\pi(k_x x + k_y y)}\, dx\, dy$
- Inverse: $m(x, y) = \iint M(k_x, k_y)\, e^{+i2\pi(k_x x + k_y y)}\, dk_x\, dk_y$

Under this convention, multiplying k-space by a phase ramp $e^{-i2\pi\, \delta x\, k_x}$ (or $e^{-i2\pi\, \delta y\, k_y}$) produces a spatial shift of $+\delta x$ in x (or $+\delta y$ in y).

Quick mapping reminders

- $M(k_x, k_y) = \mathcal{F}\{m(x, y)\}$
- $\mathrm{FOV}_x \approx \frac{1}{\Delta k_x}, \quad \mathrm{FOV}_y \approx \frac{1}{\Delta k_y}$
- Resolution and discrete-grid relations:

- $\Delta x = \frac{1}{N_x \, \Delta k_x}, \quad \Delta y = \frac{1}{N_y \, \Delta k_y}$
- $k_{\max,x} \approx \frac{N_x}{2} \, \Delta k_x, \quad k_{\max,y} \approx \frac{N_y}{2} \, \Delta k_y$
- Equivalently, $\Delta x \approx \frac{1}{2 \, k_{\max,x}}, \quad \Delta y \approx \frac{1}{2 \, k_{\max,y}}$

- PSF forms:
  - Rectangular truncation $|k_x| \leq K_x$, $|k_y| \leq K_y$ yields $\mathrm{PSF}(x, y) \propto \mathrm{sinc}(2K_x x) \, \mathrm{sinc}(2K_y y)$
  - Circular truncation $\sqrt{k_x^2 + k_y^2} \leq K$ yields isotropic Bessel blur $\mathrm{PSF}(r) \propto \frac{J_1(2\pi K r)}{r}$ with $r = \sqrt{x^2 + y^2}$
- Multiplying in k-space $\rightarrow$ convolution in image space; phase ramps in k-space $\rightarrow$ spatial shifts; coordinate warps in k-space $\rightarrow$ geometric distortions.

0.1 Discretized form

The MRI operator is discretized as follows. We will use this convention to describe the distortions and reconstruction algorithms.

$$\mathbf{y} = \mathbf{A}\mathbf{x}$$

where

$$\mathbf{A} = \mathbf{KMFS}$$

where **MFS** is the standard MRI operator consisting of sensitivity maps, FFT and mask, and $\mathbf{K}$ is a matrix representing k-space distortion.

0.2 Reconstruction

We detail linear reconstruction solvers that form the basis of more non-linear algorithms. Recall that the zero-filled reconstruction is simply the backprojection (adjoint):

$$\hat{\mathbf{x}}_{\mathrm{ZF}} = \mathbf{A}^\top \mathbf{y}$$

Note that in many cases, the k-space distortion $\mathbf{K}$ is diagonal, such that $\mathbf{K}^\top = \mathbf{K}$.

The least-squares (i.e. the pseudo-inverse), or *SENSE* reconstruction:

$$\hat{\mathbf{x}}_{\mathrm{LS}} = \arg\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 = \mathbf{A}^\dagger \mathbf{y}$$

which is solved efficiently with conjugate-gradient or LSQR. Note that in the single-coil case, this collapses to the adjoint.

0.3 Image Properties

As a reminder, the following image properties should be covered:

- **Pixel resolution / spacing (General properties)** Preservation of the ratio of pixels per content.
  Example: Low pixel resolution in reconstructed image
- **Sharpness (Glancing/Full image)** Anatomical regions and structures appear globally clear
  Example: Motion in X-Ray scan or individual anatomical layers unclearly visible in retinal OCT images
- **Intensity uniformity (Glancing/Full image)** Global uniformity of intensity levels; absence of bias fields or illumination gradients
  Example: Shading artifacts in US or dark bands in CT due to beam hardening or inhomogeneous background intensity in a microscopy image
- **Noise level (Glancing/Full image)** Random intensity fluctuations that obscure the underlying signal. Example: Rician noise in MRI
- **RoI homogeneity (Glancing/Objects of interest)** Uniform appearance of intensity of signal within main RoI that are expected to be homogeneous
  Example: Nonuniform appearance of bones in CT or nonuniform appearance of gray matter in a full MRI
- **Local signal preservation (Scanning)** Local information content is not diminished, e.g., by reconstruction artifacts; identification of hallucinated artifacts
  Example: Block or streak artifacts in MRI, near-field effects in ultrasound or vessel-like structure distorted in a denoised MR image.
- **Edges (Scanning)** Clarity and continuity of boundaries between anatomical or structural regions.
  Example: Blurred separation of grey and white matter in brain MRI or unclear vessel
- **Contrast preservation of anatomical structures (Scanning)** Subtle uniformity of intensity levels - preservation of relevant intensity differences between different objects as well as background.
  Example: Inconsistent contrast between tumor and liver parenchyma in CT or inconsistent intensity level differences between gray and white matter in T1-weighted MRI

1 Distortions

K-space masking (Resolution)

- **Isotropic resolution reduction (low-pass truncation)**
  How: keep only samples with $\sqrt{k_x^2 + k_y^2} \leq K$ (circular mask), set others to zero.
  Knob: K as a fraction of the original $k_{\max}$ (e.g., 0.3-0.8).
  Expect: global blur, loss of fine detail; resolution tracks K. The induced PSF is isotropic jinc: $\mathrm{PSF}(r) \propto \frac{J_1(2\pi K r)}{r}$.
  Image Property: Sharpness (Glancing), Edges (Scanning)

- **Anisotropic resolution change**
  How: crop differently along axes, e.g., keep $|k_x| \leq K_x$ and $|k_y| \leq K_y$ with $K_x \neq K_y$.
  Knobs: K_x, K_y.
  Expect: direction-dependent blur; elongated PSF (e.g., blur more along y if K_y smaller).
  Image Property: Sharpness (Glancing), Edges (Scanning)

- **Zero-filling (apparent resolution/pixel size change without true resolution gain)**
  How: zero-pad the k-space array to larger size in both axes before IFFT.
  Knob: pad factor (e.g., 2×, 4×).
  Expect: smaller pixels, smoother-looking edges; no true resolution improvement. **Image Property:** Pixel Resolution

- **FOV reduction and aliasing (decimate phase-encode lines)**
  How: keep every R-th ky line; equivalently increase Δk_y by factor R.
  Knob: $R \in \{2, 3, 4\}$.
  Expect: wrap-around/aliasing in y; nominal resolution from retained bandwidth unchanged; $\mathrm{FOV}'_y = \mathrm{FOV}_y/R$. **Image Property:** Local signal preservation (Scanning)

- **Variable-density bandwidth reduction (smooth taper)**
  How: multiply by a radial taper to suppress outer k-space smoothly, e.g., $H(k_x, k_y) = \exp\big(-(k_x^2 + k_y^2)/\kappa^2\big)$ or Kaiser/Hamming equivalents.
  Knob: κ, window shape.
  Expect: controlled resolution loss with less Gibbs ringing vs hard truncation. **Image Property:** Edges (Scanning), Sharpness (Glancing), ROI homogeneity (?)

Transforming the k-space (Motion)

- **Apodization (global blur and ringing suppression)**
  How: multiply by smooth windows: Gaussian, Hamming/Hann, Kaiser-Bessel.

Examples: $H(k_x, k_y) = \exp\big(-(k_x/\kappa_x)^2 - (k_y/\kappa_y)^2\big)$, or separable Hamming in each axis.
Knobs: κ_x, κ_y (or window α), choose isotropic/aniso.
Expect: reduced Gibbs ringing, softer edges.
Image Property: Sharpness (Glancing), Edges (Scanning)

- **Directional sharpness control**
  How: use different window strengths per axis, e.g., $H(k_x, k_y) = W_x(k_x)\, W_y(k_y)$ with W_x mild, W_y stronger.
  Expect: blur stronger along chosen axis; mimics T2* decay along phase-encode. **Image Property:** Sharpness (Glancing), Edges (Scanning)

- **High-frequency boost (mild sharpening)**
  How: multiply by $H(k_x, k_y) = 1 + \beta \left(\frac{\sqrt{k_x^2 + k_y^2}}{k_{\max}} \right)^p$.
  Knobs: small β (0.1-0.5), $p \geq 1$.
  Expect: crisper edges but amplified noise and potential ringing. **Image Property:** Sharpness (Glancing), Edges (Scanning), Local signal preservation (Scanning), ROI homogeneity (Glancing), Noise level (Glancing)

- **EPI N/2 ghost (odd/even readout phase mismatch)**
  How: multiply odd ky lines by $e^{i\phi}$, or apply an alternating kx phase ramp $e^{i2\pi\, \delta x\, k_x}$ to odd vs even lines.
  Knobs: ϕ (e.g., 5-30°), δx in pixels.
  Expect: ghost offset by ∼half FOV along y, plus intensity modulation. Ghost strength for the odd/even phase model decomposes as
  $$a = \tfrac{1+e^{i\phi}}{2}, \quad b = \tfrac{1-e^{i\phi}}{2},$$
  with magnitudes $|a| = \cos(\phi/2)$, $\quad |b| = \sin(\phi/2)$, and ghost-to-main ratio $\tan(\phi/2)$. The ghost is located at $\mathrm{FOV}_y/2$. **Image Property:** Local Signal Preservation (Scanning),

- **Line-by-line motion ghosts**
  Translation: for each ky line l, apply $e^{-i2\pi(k_x \Delta x_l + k_y \Delta y_l)}$ with piecewise-constant or ramping $\Delta x_l, \Delta y_l$.
  Rotation (approximate): alternate blocks of lines with opposite small translations; or regrid ky indices as if rotated trajectory.
  Knobs: max pixel shift (e.g., 0.5-3 px), pattern over lines.
  Expect: ghosts, blurring, and smear depending on motion timing.
  Image Property: Local Signal Preservation (Scanning), Sharpness (Glancing)

- **Off-resonance/field inhomogeneity (Cartesian distortion and blur)**
  How: apply per-line phase accrual tied to acquisition time: $M_l(k_x) \leftarrow M_l(k_x)\, e^{i2\pi\, \Omega_l\, t(k_x)}$, where Ω_l varies across ky (e.g., linear or sinusoidal).
  Knobs: Ω_l magnitude, functional form; $t(k_x)$ scale.
  Expect: geometric distortion and blur predominantly along phase-encode.
  Image Property: Sharpness (Glancing)

Add multiplicative Bias Field

- **Simulate Bias Field**
  How: MRI magnetic field inhomogeneity creates intensity variations of very low frequency across the whole image. The bias field is modeled as a linear combination of polynomial basis functions, as in K. Van Leemput et al., 1999, *Automated model-based tissue classification of MR images of the brain*. Default order p of the polynomial basis is 3. See also: `https://github.com/TorchIO-project/torchio/blob/main/src/torchio/transforms/augmentation/intensity/random_bias_field.py` $B(x,y) = \sum_{p_x=0}^{3} \sum_{p_y=0}^{3} \cdot x^{p_x} . p_y$.
  Then set image I to $I \cdot \frac{1}{e^{B(x,y)}}$
  Tolga:
  k-space data can be IFFT'ed into image domain, modulated via multiplication of the image-domain bias fields, and then FFT'ed back again.
  Matthias:
  I implemented this in Kspace via

  $$K^{prime} = B \cdot K$$

  where B is an appropriate weighting function (centered / non-centered, (an)isotropic gaussian with some transformation to have a certain edge gain etc).

  I hope this is ok as well. **Knob:** maximum amplitude A of polynomial function, so $c_{(p_x,p_y)}$ are randomly sampled from a uniform distribution in $(0, A)$
  Expect: Smooth background gradient across entire image **Image Property: Intensity uniformity (Glancing/Full image)**

Adding Noise

- **Adding Gaussian Noise**
  How: For each pixel, sample noise to be added from a Gaussian (Normal) distribution

  $$K' = K + n,\ n \sim N(\mu = 0, \sigma)$$

  Knob: σ Standard deviation of Gaussian distribution
  Expect: Increased Gaussian noise in the image Image Property: Noise Level (Glancing)

- **Adding Rician Noise**
  How: For each pixel, sample noise to be added from a Rician distribution, see https://deepinv.github.io/deepinv/api/stubs/deepinv.physics.RicianNoise.html

  $$y = \sqrt{(x + \sigma\epsilon_1)^2 + (\sigma\epsilon_2)^2},\ \epsilon_1 \sim N(0, I),\ \epsilon_2 \sim N(0, I)$$

  Knob: σ Standard deviation of Gaussian distribution
  Expect: Increased Gaussian noise in the image Image Property: Noise Level (Glancing)

2 Other

- **Modification of sensitivity map for multi-coil Acquisitions**
  How: Many ways to do this, but one thing could be to pass the magnitude of the sensitivities through a compressive or expansive nonlinear transformation (e.g., log or exp).
  Expect:
  Knob:
  Image Property: RoI Homogeneity (Glancing)

Removed

- **Unsharp masking in k-space**
  How: compute a low-pass $L(k_x, k_y)$ (e.g., Gaussian), then apply $M'(k_x, k_y) = M(k_x, k_y)\left[1 + \beta\left(1 - L(k_x, k_y)\right)\right]$.
  Knobs: β (0.1-0.3), low-pass width.
  Expect: controlled sharpening with less DC/low-frequency inflation. **Image Property:** Sharpness (Glancing), Edges (Scanning)
  Removed: Redundant with high frequency boost

- **Regularized inverse of known blur**
  How: if blur window $L(k_x, k_y)$ is known, apply $H(k_x, k_y) = \frac{L^*(k_x,k_y)}{|L(k_x,k_y)|^2+\lambda}$ (Wiener-like).
  Knob: λ.
  Expect: sharpness restoration and noise amplification trade-off. **Image Property:** Sharpness (Glancing), Edges (Scanning), Noise level (Glancing) **Removed:** Redundant with high frequency boost, Tolga: we may choose to assume that the blur kernel is known, and then compensate for that only, or just assume that it is unknown and pick a popular high-pass filter.

- **Gibbs ringing**
  How: hard truncate with rectangular mask: keep $|k_x| \leq K_x$, $|k_y| \leq K_y$.
  Knobs: K_x, K_y.
  Expect: oscillations near sharp edges; stronger with harder crop. **Image Property:** Local Signal Preservation (Scanning), Roi Homogeneity (Glancing) **Removed:** Redundant with isotropic/anisotropic resolution reduction

- **Aliasing (wrap-around) from undersampling ky**
  How: skip ky lines by factor R without parallel imaging.
  Knob: R.
  Expect: replicated objects along y spaced by FOV_y/R.

Image Property: Local Signal Preservation (Scanning)
Removed: Redundant with FOV reduction and aliasing.

- **Chemical shift along readout for multi-echo Dixon separation**
  Model: For echo e with echo time TE_e, the fat component accrues phase from the species offset both across TE and during the readout. In discrete k-space (per echo e, ky line l, and readout index j):

  $$M_e[j,l] \;=\; M_{\text{water}}[j,l] + M_{\text{fat}}[j,l]\;\exp\!\big(i2\pi\,\Delta f\,TE_e\big)\;\exp\!\Big(i2\pi\,\delta x_{\text{pix}}\,\frac{j-j_0}{N_x}\Big)$$

  Here j_0 is the echo-center sample index and N_x is the number of readout points. The kx-dependent phase is equivalent to a shift of the fat image along x. **Knobs:** species amplitude ratio α (e.g., 0.2-0.6), frequency offset Δf, and the derived pixel shift $\delta x_{\text{pix}} \;\approx\; \Delta f\,\frac{N_x}{\text{BW}_{\text{ro}}}$ with BW_{ro} the readout bandwidth (Hz). The sign of δx_{pix} depends on readout polarity. **Expect:** Across echoes, fat–water relative phase rotates by $e^{i2\pi\Delta f\,TE_e}$ (enabling Dixon separation). Within each echo, the fat component appears shifted along x by δx_{pix}, producing chemical-shift misregistration unless corrected. **Image Property:** Local Signal Preservation (Scanning)

- **Zipper/stripe artifact from k-space spike**
  How: add a spike at index (k_{x0}, k_{y0}): $M'(k_x,k_y) = M(k_x,k_y) + A\,\delta(k_x - k_{x0})\,\delta(k_y - k_{y0})$.
  Knobs: amplitude A, location (k_{x0}, k_{y0}).
  Expect: periodic stripes; orientation depends on spike location. **Image Property:**

- **Gradient delay or ADC timing error (even/odd readout asymmetry)**
  How: shift odd ky lines along kx via phase ramp: $M_{\text{odd}}(k_x) \leftarrow M_{\text{odd}}(k_x)\,e^{-i2\pi\,\Delta x\,k_x}$; optionally use a small Δx varying with ky.
  Knobs: Δx (0.1-1 px).
  Expect: blurring, N/2-like ghosts, small geometric shifts. **Image Property:** Sharpness (Glancing), Local Signal Preservation (Scanning)

- **Partial Fourier (half-scan) artifacts**
  How: zero one side of ky (e.g., negative ky), then fill via imperfect conjugate symmetry (add phase errors).
  Knobs: fraction of ky kept (50-70%), phase error magnitude.
  Expect: blurring, ringing, and phase-related ghosting. **Image Property:**

- **Phase-encode scaling/offset (gradient miscalibration)**
  How: rescale ky indices $k'_y = \alpha\,k_y + d$ and reconstruct as if unscaled.
  Knobs: α (0.95-1.05), d (offset in samples).
  Expect: y-scaling and shift; can create slight shear with d varying across k_x.

Image Property: ?
Removed: No relevant property affected

- **Random line-wise phase (system instabilities)**
  How: multiply each ky line by $e^{i\phi_l}$ with ϕ_l drawn from a small-noise distribution.
  Knob: phase std dev (e.g., 5-20°).
  Expect: incoherent ghosts and mild blur.

Coordinate scaling (simulate gradient scaling / bandwidth change)
How: resample onto scaled coordinates $(k'_x, k'_y) = (\alpha_x k_x, \alpha_y k_y)$ and reconstruct on the original grid.
Knobs: α_x, α_y close to 1 (e.g., 0.9-1.1).
Expect: continuous-domain outcome $m_{\text{out}}(x, y) = \frac{1}{\alpha_x\,\alpha_y}\, m\Big(\frac{x}{\alpha_x}, \frac{y}{\alpha_y}\Big)$,
i.e., structures are magnified by α_x along x and α_y along y (with intensity scaled by $1/(\alpha_x\alpha_y)$ in the continuous model). Interpolation may add minor artifacts.
Image Property: Pixel Resolution
Removed: Already striked out

3 Reconstruction algorithms

TODO list standard algorithms from DeepInverse:

1. Optimisation algorithms (e.g. FISTA, ADMM, etc.) regularised with various regularisers (TV, wavelets, etc.)
2. PnP/RED i.e. model-based deep learning optimisation approaches with pretrained denoisers
3. Pretrained unrolled algorithms e.g. MoDL, (VarNet) etc.
4. Pretrained feedforward models and foundation model i.e. RAM
5. Posterior sampling algorithms (diffusion sampling, MCMC) with pretrained denoisers.